

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2012

CL308 Geotechnical Engineering-II

Date: 09.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain standard penetration test in detail. Also explain corrections for the N-value. [05]
(b) List out the in-situ tests for soil exploration. [02]
- Q - 2 (a) Enlist the different forces acting on the well foundation. Explain any three in detail. [07]
(b) Write a short note on grip length. [03]
(c) List out the criterias on which lateral earth pressure depends. [04]

OR

- Q - 2 (a) Draw different shapes of wells & give their significance. [05]
(b) Explain well foundation in detail & show all parts of it with neat sketch. [05]
(c) Define lateral earth pressure. Explain its types in detail. [04]
- Q - 3 (a) A retaining wall having 5m height supports a backfill having $C = 20\text{kN/m}^2$, $\phi = 30^\circ$ [05]
and $\gamma = 20\text{kN/m}^3$ with horizontal top flushed with the top of wall. The backfill carries
a surface load of 25kN/m^2 . If the wall is pushed towards the backfill, compute the
total passive pressure on the wall & its point of application.
- (b) A concentrated load of 100 kN acts on the surface of soil. Plot the following: [05]
(i) Variation of vertical stress increment due to load on horizontal plane at a depth of
3m up to a horizontal distance of 3 m on either side of the centre (ii) Variation of
horizontal stress increment due to load on vertical plane at a distance of 3m up to a
depth of 2 m (vertical intervals = 0.50m). Also plot the graph.

Depth 'z'	0.5	1.0	1.5	2.0
Influence factor ' I_B '	0.0085	0.0844	0.1904	0.2733

- (c) Explain in detail the equivalent point load method of stress distribution. [04]

OR

- Q - 3 (a) For the retaining wall shown in Figure 1, assume the active condition. Using [05]
Rankine's active earth pressure theory determine:

- (i) Active force per meter of wall
- (ii) The location of the resultant line of action ($C = 0$).
- (b) A raft of size $4\text{m} \times 4\text{m}$ carries a load of 200 kN/m^2 . Determine vertical stress increment at a point 4m below the centre of the loaded area using Boussinesq's theory. Compare the results with that of obtained by equivalent point load method (for 1 equivalent point load & 4 equivalent point loads). [05]
- (c) What are the assumptions made in Boussinesq's theory? Write the equation for vertical pressure under a uniformly loaded circular area. [04]

SECTION – II

- Q - 4 (a) Explain $C-\Phi$ analysis for stability of slopes by Swedish circle method. [04]
- (b) Explain different factor of safety used in stability analysis of slopes. [03]
- Q - 5 (a) A strip footing of 2 m width and 1.5 m depth is resting on cohesionless soil having unit weight 19 kN/m^3 . The shear strength parameters are $C=0$ and $\Phi=35^\circ$. Determine ultimate bearing capacity in the following cases: [07]
- (i) Water table is 2.5m below G.L.
- (ii) Water table is 0.5m below G.L. Consider general shear failure in both cases.
- Take $N_c = 57.8$; $N_q = 41.4$; $N_\gamma = 42.4$.
- (b) List the factors affecting bearing capacity and discuss the effect of water table on bearing capacity. [05]
- (c) Write two dynamic formulae to determine pile load capacity. [02]

OR

- Q - 5 (a) A wall footing 1m wide and 2m deep is to be placed on sandy soil strata. The site investigation gives the following SPT values: [07]

Depth from G.L.	1.5	2.25	3.0	3.75
SPT values (N)	10	12	14	15

Using $\gamma=18 \text{ kN/m}^3$, factor of safety=2.0, permissible settlement=30mm, water table is 2.5m below G.L.; calculate allowable pressure for the footing.

- (b) Write assumptions made in the theory of Terzaghi's analysis. Give bearing capacity equations for strip, circular and square footing as per Terzaghi's theory. [05]
- (c) Distinguish between end bearing pile and friction pile. [02]

- Q - 6 (a) An embankment is constructed by sandy clay having cohesion of 30 kN/m^2 , angle of internal friction is 20° and unit weight of 20 kN/m^3 . The height of embankment is 10m , $\alpha=25^\circ$ & $\beta=15^\circ$. Determine the factor of safety using method of slices [05]

considering toe failure.

- (b) Explain group action of piles. [03]
- (c) A precast concrete pile 300mm diameter and 10 m long is driven in to sand deposit by single acting steam hammer. The weight of hammer is 12 kN and stroke is 750mm. The average penetration in the last 5 blows is 4mm. Estimate the ultimate bearing capacity of the pile based on Hiley's formula. Take $C=0.005\text{m}$ & $e = 0.5$. [06]

OR

- Q - 6 (a) List various methods of improving stability of slopes. [02]
- (b) A long natural slope of cohesion less soil is inclined at 15° to the horizontal. Taking $\phi=25^\circ$, determine the F.O.S. of the slope. If slope is completely submerged, justify your answer for change in F.O.S.? [03]
- (c) Define negative skin friction. How it affect the pile load carrying capacity? [03]
- (d) A group of 9 piles, 9 m long is used as the foundation of column. The piles are 30 cm in diameter with centre to centre spacing 90 cm. the subsoil consists of clay with cohesion 85 kN/m^2 . Estimate safe load. Take F.O.S. = 3. [06]

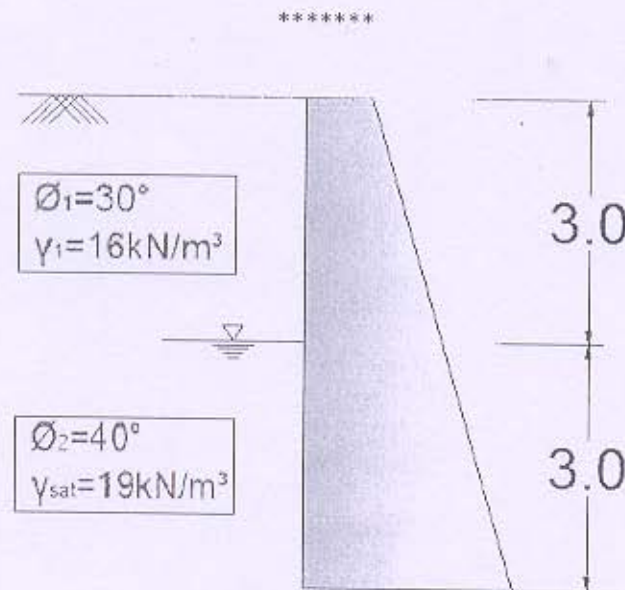


Figure 1